

Global CO₂ Emissions: Who Emits, How Much, and What's Changing

Fossil fuels & industry, 1750–2024

Audience: Policy analysts, climate negotiators, sustainability researchers

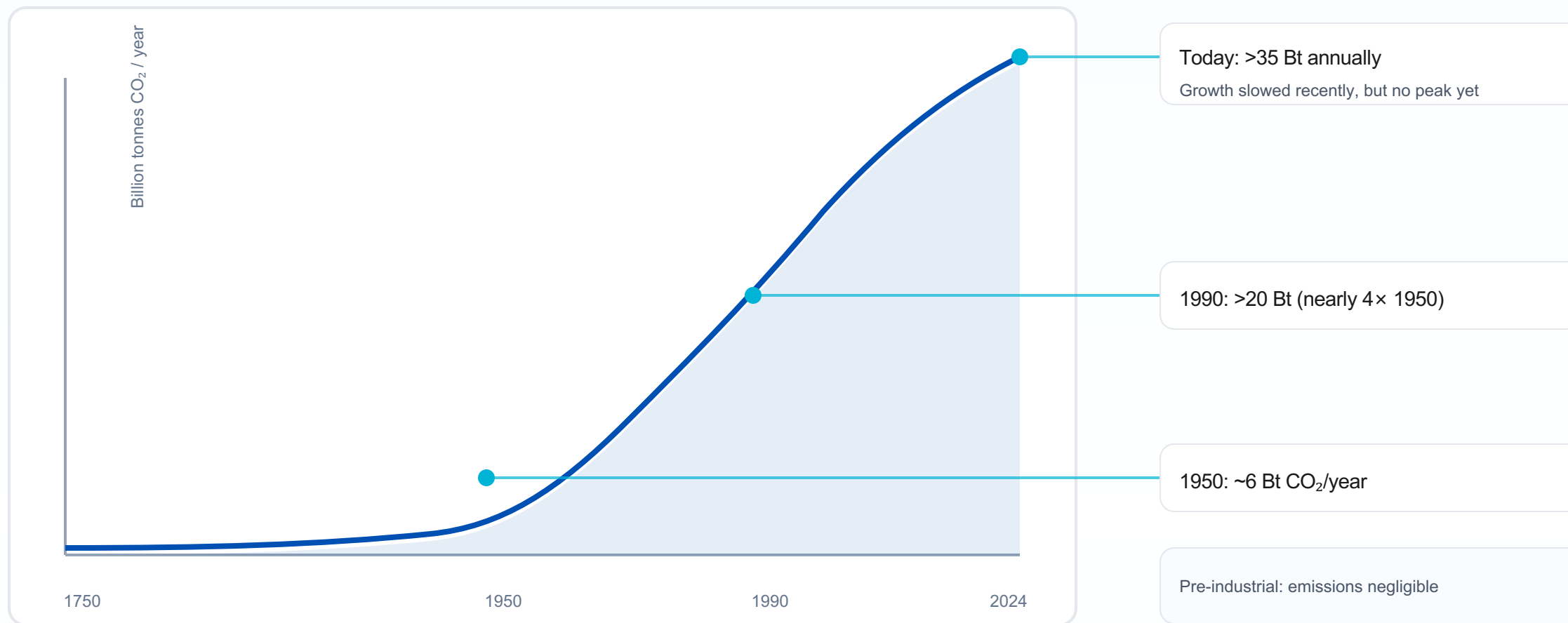
Data: Global Carbon Budget (2025) via Our World in Data

Updated: 2026-04-21

2 Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

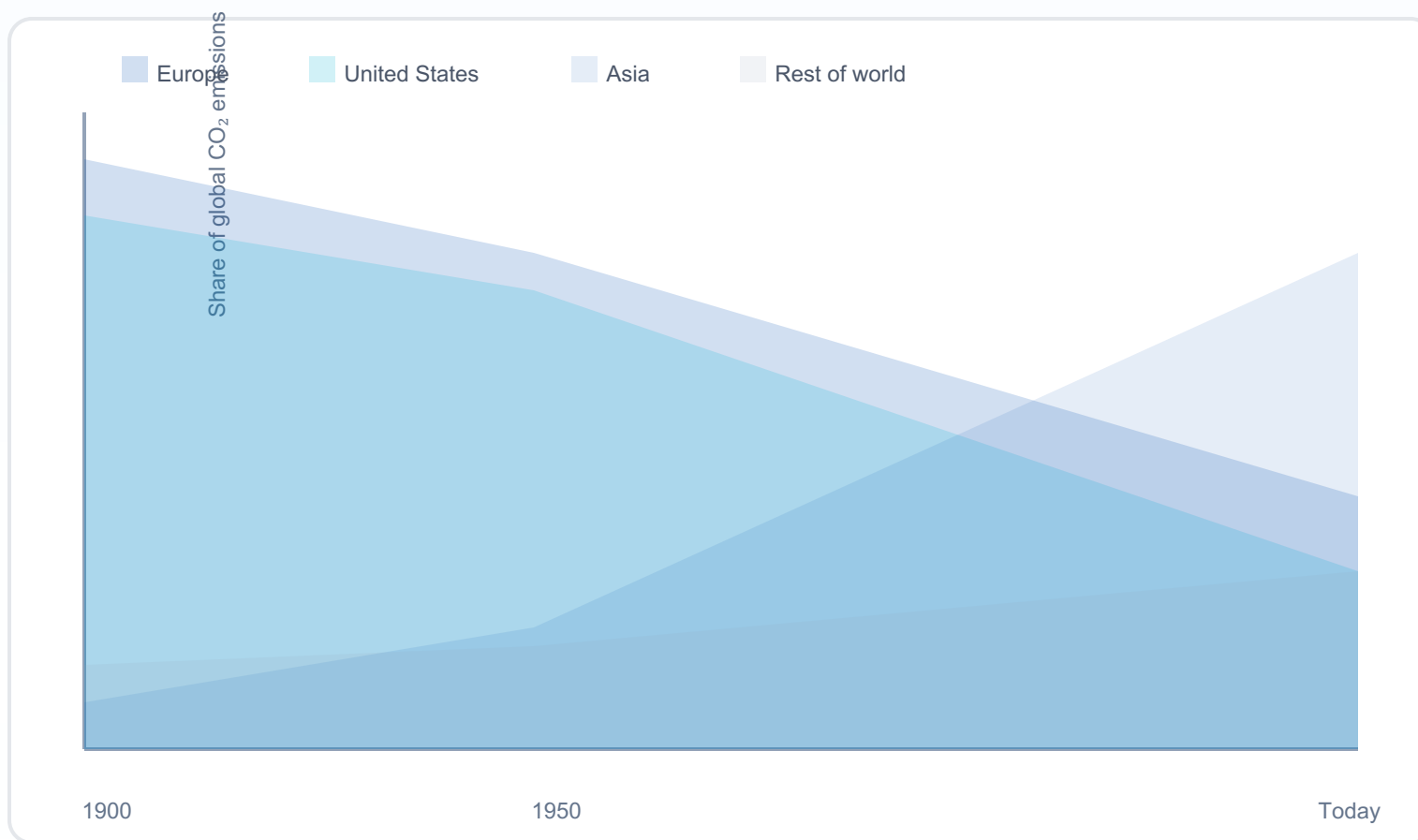
Global Carbon Budget (2025) via Our World in Data

Takeaway: Emissions surged after 1950; growth slowed recently, but no peak yet.



Note: Curve shown for direction only; labeled values are from the provided source brief.

Takeaway: The center of emissions shifted from Europe/US to Asia (led by China).



Key share shifts (from brief)

- 1900: Europe + US > 90%
- 1950: Europe + US > 85%
- Today: Europe + US < one-third
- Asia now ~half (led by China)

Visual note

Stacked areas are schematic.

Only the labeled statements are quantitative.

Reference image: figure_6.jpg (OWID).

Takeaway: In 2024, China's share (~31%) is larger than the US + EU-27 combined (~19%).

China
~31%

Rising, growth slowing

United States
~13%

Declining

India
~8%

Rising

EU-27
~6%

Declining

Russia
~5%

Stable

Africa
~3–4%

S. America
~3–4%

Int'l aviation & shipping
~3%

What to notice

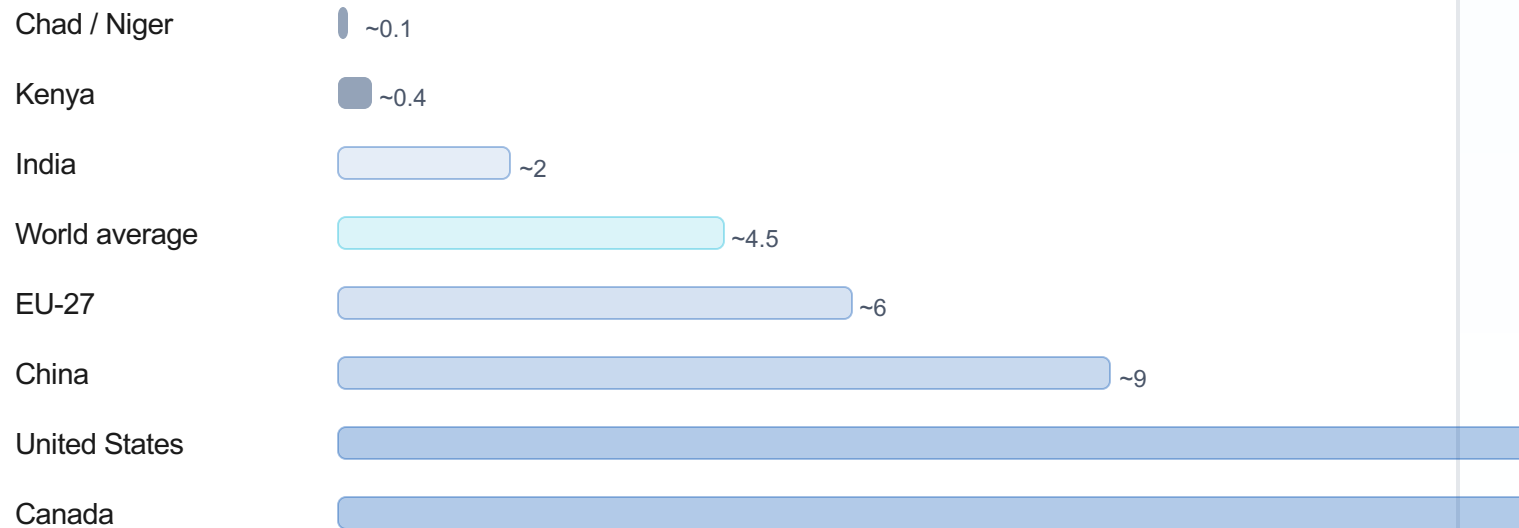
- China accounts for ~31% of 2024 emissions.
- US (~13%) is declining; India (~8%) rising.
- Africa and South America are each ~3–4%.
- Int'l aviation & shipping is ~3%.

Comparison (from brief):

Africa and South America each comparable to international aviation & shipping combined.

Takeaway: A roughly 150× per-capita gap persists between lowest and highest emitters.

tCO₂ per person (approx., 2024)



150× gap (from brief)

Lowest: ~0.1 t/person

Highest large-country: ~14–15 t/person

Interpretation

Per-capita emissions reframe responsibility:

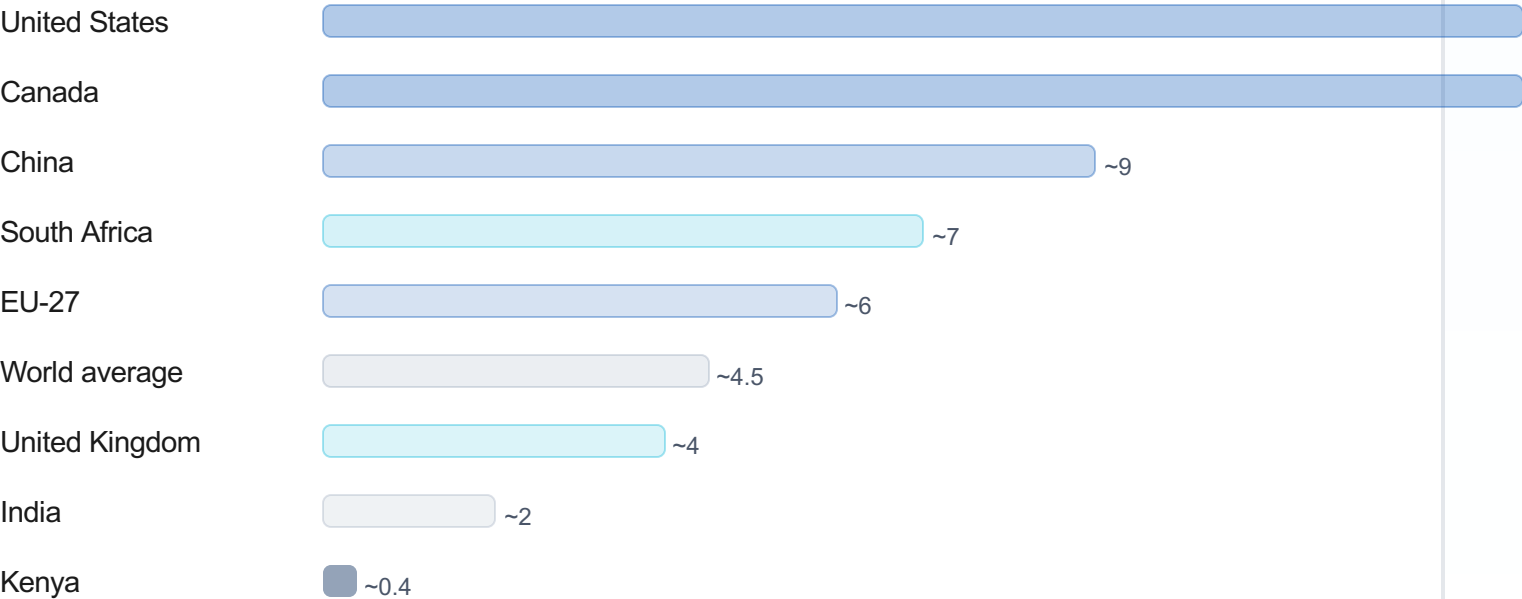
- Many nations remain near-zero contributors.
- Some wealthy nations are much lower (e.g., UK vs US) despite similar living standards.

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

Global Carbon Budget (2025) via Our World in Data

Takeaway: China is the largest total emitter, but per-capita emissions (~9 t) remain below the US (~14 t).

tCO₂ per person (approx., 2024)

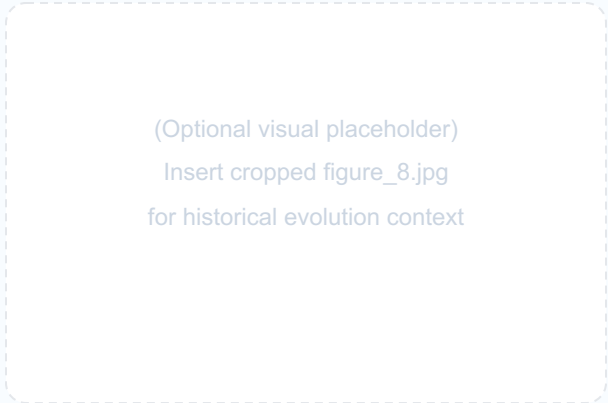


Interpretation

- US and Canada are ~3× the world average.
- China’ s per-capita (~9) is below US (~14).
- India remains very low at ~2 t/person.

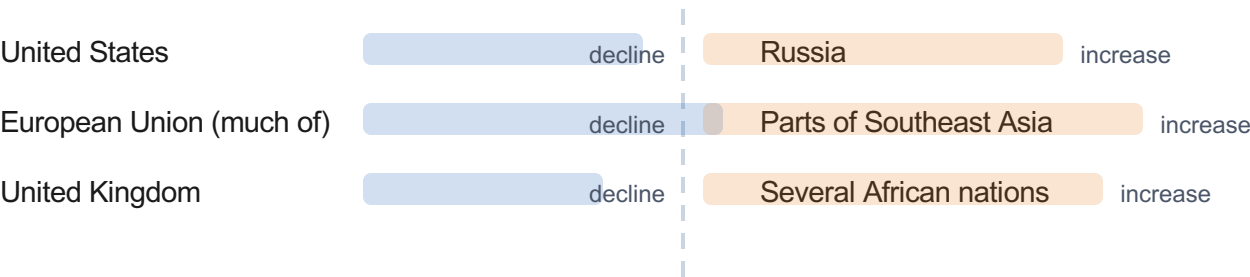
Context

Figure reference: figure_8.jpg (per-capita over time).



Takeaway: 2024 was mixed—declines in many Western nations, increases across parts of Asia and Africa.

Annual % change in CO₂ emissions (2024) — qualitative bins

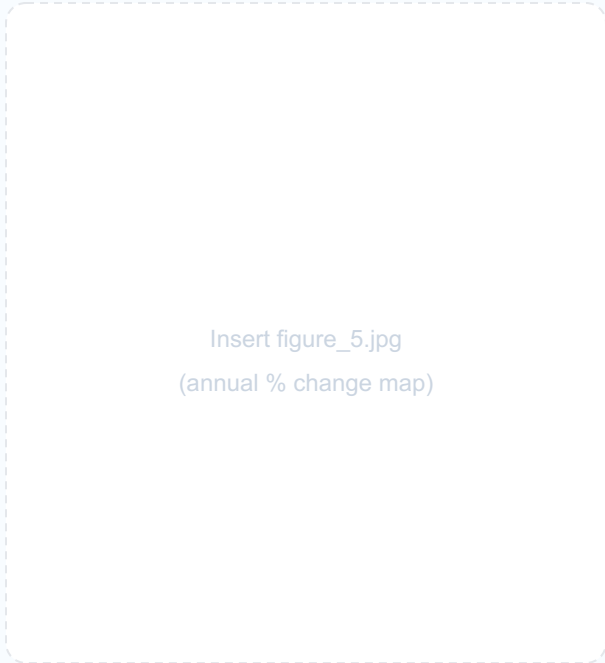


Notable from brief:

Some Central Asia (Turkmenistan/Uzbekistan region) saw increases exceeding 10%.

0%

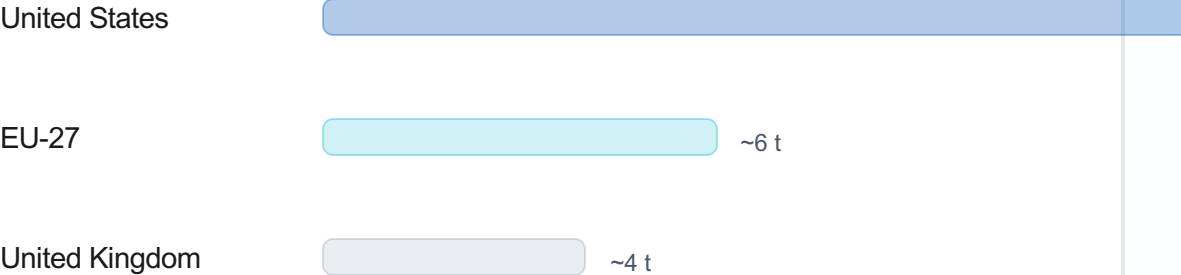
Map reference



Blue = decline, orange/red = increase

Takeaway: Similar prosperity can mean very different per-capita emissions—energy choices matter.

Per-capita CO₂ (approx., 2024)



Why such differences?

From the source brief:

Energy mix choices can lower emissions

- France, the UK, and Portugal have much lower per-capita emissions than Germany, the Netherlands, or the US.
- A key driver is higher nuclear + renewables vs. fossil-heavy power.

(Optional placeholder)

Add energy-mix illustration
without adding new numeric shares

What the data imply for policy debates

- Global emissions exceed 35 Bt/yr and haven't peaked.
- China is the largest annual emitter (~31%), but per capita (~9 t) is below the US (~14 t).
- Historical responsibility shifted: Europe + US >90% (1900) → < one-third today.
- Per-capita gaps of ~150× persist: lowest ~0.1 t vs high emitters ~14–15 t.
- Energy policy and technology choices can decouple prosperity from emissions.

Anchor facts (from brief)

Global emissions

>35 Bt/yr

China share (2024)

~31%

Per-capita range

~0.1 t → ~14 t

Next step: use both totals and per-capita views in negotiation framing.